



HR Analytics: Workforce Performance, Risk Assessment, and Strategic Recommendations

Transforming Employee
Data into Predictive
Retention Insights

NHA-ALX3-DAT1-G3

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Project Overview & Objectives

To move HR from reactive reporting to **proactive, data-driven decision making**.

1. Analyze historical workforce performance and demographics.
 2. Identify operational gaps in management and training.
 3. **Predict** future employee attrition using Machine Learning.
- Analysis of 1,470 employees spanning the years 2013–2022.
 - List of the 5 key objectives or a simple icon set representing Data -> Insight -> Action.

Executive Summary

- Current Attrition is 16.12%, but 14% of the Workforce is at Immediate Risk.
- Key Metrics:
 - Overall Attrition Rate: 16.12% (Historical Baseline).
 - High-Risk Headcount: 208 Employees (Identified by ML Model).
 - Rating Alignment Gap: 0.51 (Self vs. Manager Perception gap).



Predictive Risk Assessment

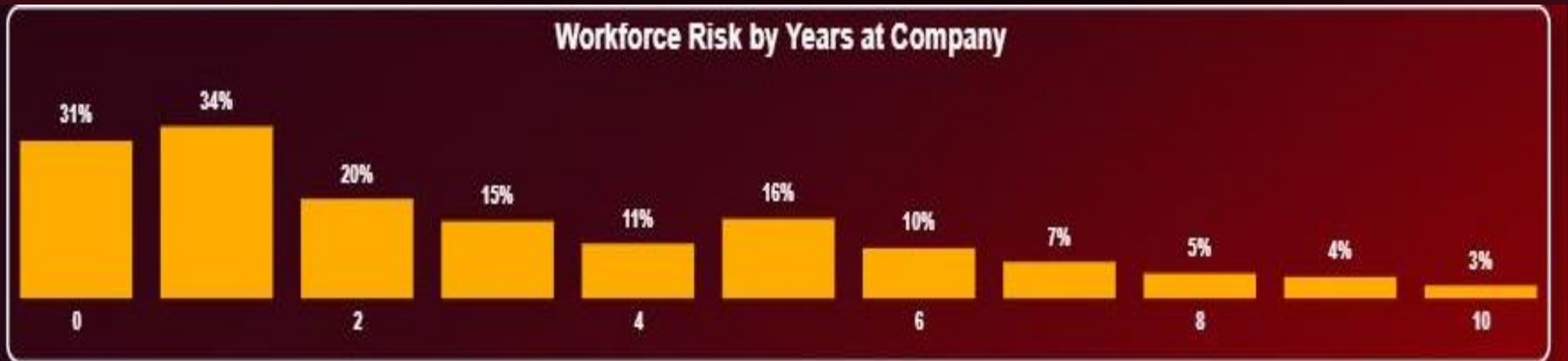
- The predictive model has identified specific "High-Risk" employees (Probability $\geq 80\%$)
- **Risk by Department:**
 - **Sales:** 20% Avg Risk (Highest).
 - **Human Resources:** 19% Avg Risk.
 - **Technology:** 13% Avg Risk.
- **Strategic Implication:** Retention efforts must be targeted, not generic.



Critical Vulnerability: Early Tenure

The "Danger Zone" is the First 2 Years.

- Both historical data and predictive modeling point to the same failure point.
 - **Year 0 (New Hires):** 31% Attrition Risk.
 - **Year 1:** 34% Attrition Risk.
- **Conclusion:** The organization is failing to engage employees during the onboarding and initial development phase.



Operational & Performance Gaps

Management Alignment & Training Issues.

- **Rating Disconnect:**

- **Avg Self Rating:** 3.98 vs. Avg Manager Rating: 3.47.
- **Gap (0.51):** Employees think they are performing better than managers do, leading to review-time dissatisfaction.

- **Training Decline:**

- Training utilization has dropped from a peak of 55.9% (2014) to 46.5% (2022).



Strategic Recommendations

Targeted Retention Intervention:

- HR to pull the list of **Top 208 High-Risk Employees**.
- Conduct "Stay Interviews" specifically for Sales and HR departments.

2. Manager Calibration Training:

- Launch training to fix the **0.51 Rating Gap**. Managers must communicate feedback before the formal review.

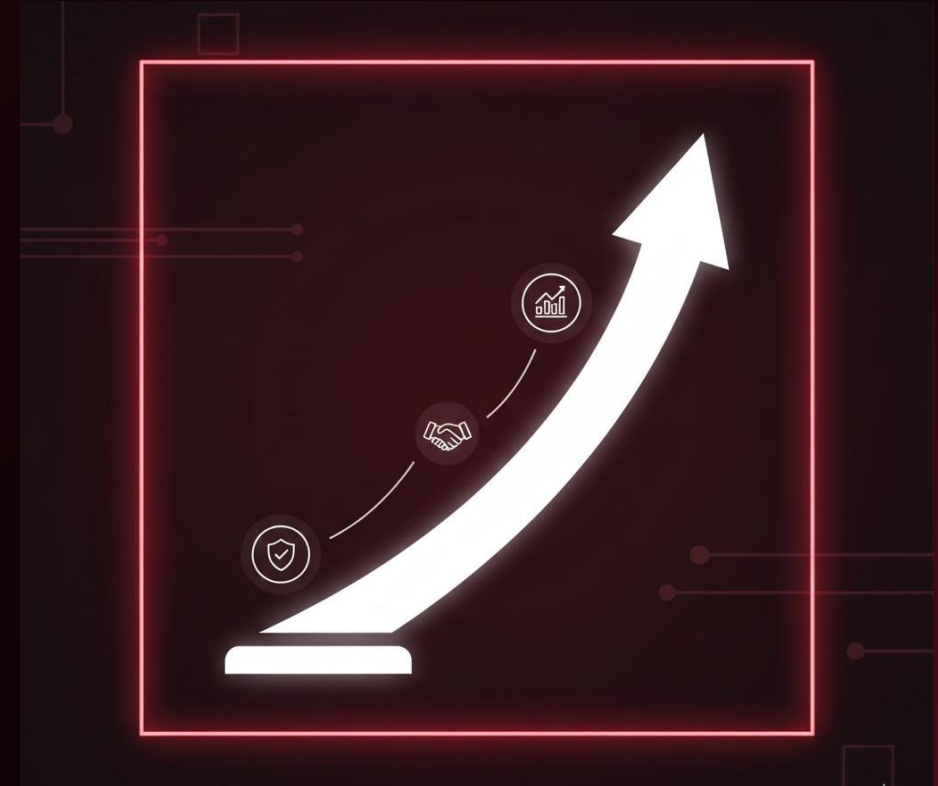
3. Revamp Onboarding (0-2 Years):

- Implement 6-month and 12-month check-ins to stop the **34% early attrition bleed**.



Impact & Next Steps

- Expected Impact:
 - Reduction in early-tenure turnover.
 - Improved manager-employee relationship scores.
 - Data-driven allocation of training budget.
- Next Steps:
 - Deploy dashboard to Department Heads.
 - Begin screening of High-Risk list immediately.



Thank You

Questions & Discussion